Mohammad Sadegh Sirjani

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Education

Ph.D., Computer Science Jan. 2025 - Present
University of Texas at San Antonio, GPA: 3.93/4.0

B.Sc., Computer Engineering Sep. 2018 - Feb. 2024
Ferdowsi University of Mashhad, Cumulative GPA: 3.02/4.0 (16.52/20), Last 60 Credits: 3.89/4.0

Research Interests

Internet of Things	Tiny AI	Edge AI
Embedded Systems	Energy Harvesting	Intermittent Computing

Research Experience

Research Assistant Jan. 2025 - Present
ASIC Lab, University of Texas at San Antonio

- Developing energy-efficient neural network accelerators for edge AI applications on IoT devices.
- Investigating ultra-low-power ML inference using hardware-software co-design approaches.

Research Assistant Aug. 2022 - Jan. 2024
Software Quality Lab, Ferdowsi University of Mashhad

- Analyzed Java application execution traces using XRebel and JRebel for software enhancement.
- Developed code analysis techniques to identify class, interface, and enum connections.
- Built tool for recommending microservice migration strategies from monolithic codebases.

Research Assistant Dec. 2023 - Aug. 2024
Web Technology Lab, Ferdowsi University of Mashhad

- Designed database structure and web-based API for psychological survey data collection.
- Integrated LLMs to generate enhanced client assessment reports from survey data.
- Successfully deployed production application through iterative prompt engineering refinements.

Publications

1. Ahmad, M., **Sirjani, M. S.**, Wei, W., & Xie, M. (2026). "MOSAIC: Multi-Domain Sensing Architecture for Efficient Computing in Domain Wall Memory". *Under review at the 44th IEEE International Conference on Computer Design (ICCD 2026).*
2. **Sirjani, M. S.**, Ahmad, M., Wei, W., Dutton, P., Yanez, D., Pan, C., & Xie, M. (2026). "SparseBeat: Adaptive Compressed Sensing for Arrhythmia Classification on Energy-Harvesting Wearables". *Under review at the ACM/IEEE International Conference on Embedded Artificial Intelligence and Sensing Systems (SenSys 2027).*
3. Mousavi, A., Nourbakhsh, E., **Sirjani, M. S.**, Slavin, R., Xie, M., Neely, L., Davis, J., & Quarles, J. (2026). "CogAdapt: Transferring Clinical ECG Foundation Models to Wearable Cognitive Load Assessment via Lead Adaptation". *Under review at the IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI 2026).* `https://arxiv.org/abs/2605.22774`.

4. Mousavi, A., **Sirjani, M. S.**, Nourbakhsh, E., Xie, M., Slavin, R., Neely, L., Davis, J., & Quarles, J. (2026). "MambaGaze: Bidirectional Mamba with Explicit Missing Data Modeling for Cognitive Load Assessment from Eye-Gaze Tracking Data". *Under review at the IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI 2026).* `https://arxiv.org/abs/2605.22775`.
5. Nourbakhsh, E., **Sirjani, M. S.**, Mousavi, A., Nguyen, K., Quarles, J., Xie, M., & Slavin, R. (2026). "Are LLM Benchmarks Already Contaminated? A Systematic Review of Contamination Detection Methods". In Proceedings of the *GEM Workshop at the 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026).* `https://doi.org/10.18653/v1/2026.gem-main.50`.
6. Ghaffari, A., Firoozi, V., Maleki, A., **Sirjani, M. S.**, & Abedini Bagha, M. (2025). "QTE-IoT: Q-Learning-Based Task Scheduling Scheme to Enhance Energy Consumption and QoS in IoT Environments". *Sustainable Computing: Informatics and Systems.* `https://doi.org/10.1016/j.suscom.2025.101247`.
7. **Sirjani, M. S.**, Maleki, A., Pakmehr, A., Abedini Bagha, M., Ghaffari, A., & Pour Haji Kazem, A. A. (2025). "Controller placement in software-defined networks using reinforcement learning and metaheuristics". *Cluster Computing*, 28(10), 660. `https://doi.org/10.1007/s10586-025-05331-y`.
8. **Sirjani, M. S.**, Ahmad, M., Mousavi, A., Nourbakhsh, E., & Nguyen, K. (2026). "Optimizing Task Scheduling in Fog Computing with Deadline Awareness". In Proceedings of the *2nd IEEE International Conference on Secure IoT, Assured and Trusted Computing (SATC 2026).* IEEE. `https://doi.org/10.1109/SATC69565.2026.11542230`.
9. **Sirjani, M. S.**, Mousavi, S. A., & Sadeghi, M. (2024). "Data mining and cloud computing for customer pattern analysis". In Proceedings of the *IEEE 10th International Conference on Artificial Intelligence and Robotics (ICAR 2024)* (pp. 339-344). IEEE. `https://doi.org/10.1109/QICAR61538.2024.10496623`.
10. Mousavi, S. A., **Sirjani, M. S.**, Bozorg Zadeh Razavi, S. J., & Nikooghadam, M. (2023). "SecVanet: provably secure authentication protocol for sending emergency events in VANET". In Proceedings of the *IEEE 10th International Conference on Information and Knowledge Technology (IKT 2023)* (pp. 86-91). IEEE. `https://doi.org/10.1109/IKT62039.2023.10433027`.
11. Mousavi, S. A., Sadeghi, M., & **Sirjani, M. S.** (2023). "A comparative evaluation of machine learning algorithms for IDS in IoT network". In Proceedings of the *IEEE 10th International Conference on Information and Knowledge Technology (IKT 2023)* (pp. 168-174). IEEE. `https://doi.org/10.1109/IKT62039.2023.10433047`.

Awards & Honors

- Winner, 2-Minute Presentation Award Competition – 63rd Design Automation Conference (DAC), 2026
- Graduate School Academic Travel Fund – University of Texas at San Antonio, 2026
- Selected as a DAC Young Fellow – 63rd Design Automation Conference (DAC), 2026
- Fan Favorite Award – Draper Data Science Business Plan Competition, 2026
- Scholarship recipient – DAiR3 Data Science Summer School, 2026
- Winner, 2-Minute Presentation Award Competition – 62nd Design Automation Conference (DAC), 2025
- Selected as a DAC Young Fellow – 62nd Design Automation Conference (DAC), 2025

Professional Service

- Reviewer – Great Lakes Symposium on VLSI (GLSVLSI), 2026

Teaching Experience

- **Fundamentals of Operating Systems** (Fall 2025) – Dr. Sam Silvestro, UTSA
- **Data Science** (Summer 2025) – Dr. Amin Sahba, UTSA
- **Computer Organization** (Summer 2025) – Dr. Subhasish Das, UTSA
- **Fundamentals of Operating Systems** (Spring 2025) – Dr. Mimi Xie, UTSA
- **Fundamentals of Cloud Computing** (Spring 2024) – Dr. Somayeh Sobati-Moghadam, FUM
- **Principles of Compiler Design** (Fall 2023) – Dr. Haleh Amintoosi, FUM
- **Fundamentals of Data Mining** (Fall 2023) – Dr. Behshid Behkamal, FUM
- **Fundamentals of Cloud Computing** (Fall 2023) – Dr. Somayeh Sobati-Moghadam, FUM
- **Principles of Compiler Design** (Spring 2023) – Dr. Haleh Amintoosi, FUM
- **Principles of Web Development** (Spring 2023) – Dr. Somayeh Sobati-Moghadam, FUM
- **Principles of Web Development** (Fall 2022) – Dr. Somayeh Sobati-Moghadam, FUM

Technical Skills

- **Programming Languages:** Python, C, C++, C#
- **ML/DL Frameworks:** TensorFlow, PyTorch, Keras, scikit-learn
- **IoT Protocols:** MQTT
- **Embedded Platforms:** NVIDIA Jetson (Orin Nano, Orin NX, AGX Orin, Nano, Xavier), Raspberry Pi 5, Arduino, STM32 ARM Cortex M4, MSP430FR5994
- **Tools & Technologies:** Docker, Kubernetes, Git, CUDA